On applying sign convention,

$$\frac{-h'}{h} = \frac{v}{-u}$$

Substituting this in the equation (6.65) for magnification,

$m = \frac{-h'}{h} = \frac{v}{-u}$; After rearranging,

$$m = \frac{h'}{h} = \frac{v}{u} \qquad (6.66)$$

The magnification is negative for real image and positive for virtual image. In the case of a concave lens, the magnification is always positive and less than one.

We can also have the other forms of equations for magnification by combining the lens equation as,

$$m = \frac{h'}{h} = \frac{f}{f+u} \text{ (or) } m = \frac{h'}{h} = \frac{f-v}{f} \qquad (6.67)$$

EXAMPLE 6.13

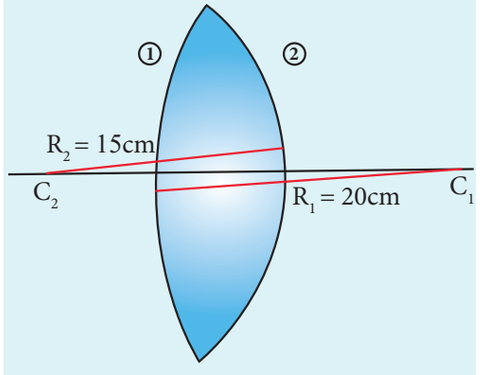
A biconvex lens has radii of curvature 20 cm and 15 cm for the two curved sufaces. The refractive index of the material of the lens is 1.5.

(a) What is its focal length?

(b) Will the focal length change if the lens is flipped by the side?

Solution

For a biconvex lens, radius of curvature of the first surface is positive and that of the second surface is negative as shown in figure.



Given, n = 1.5, R_1=20 cm and R_2= –15 cm

(a) Lensmaker's formula, $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

Substituting the values,

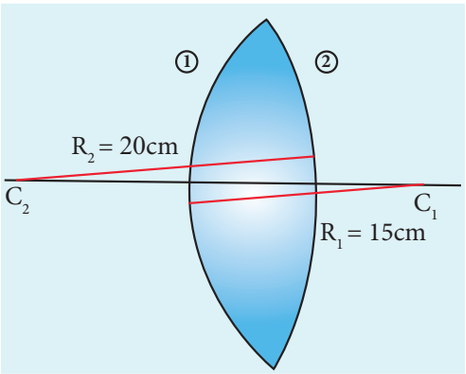
$$\frac{1}{f} = (1.5-1)\left(\frac{1}{20} - \frac{1}{-15}\right) = (1.5-1)\left(\frac{1}{20} + \frac{1}{15}\right)$$

$$\frac{1}{f} = (0.5)\left(\frac{1}{20} + \frac{1}{15}\right) = (0.5)\left(\frac{3+4}{60}\right) = \left(\frac{1}{2} \times \frac{7}{60}\right) = \frac{7}{120}$$

$$f = \frac{120}{7} = 17.14\,\text{cm}$$

As the focal length is positive the lens is a converging lens.

(b) When the lens is flipped by the side,



Now, n = 1.5, R_1= 15 cm and R_2= –20 cm

Substituting the values in the lens maker's formula,

$$\frac{1}{f} = (1.5-1)\left(\frac{1}{15} - \frac{1}{-20}\right)$$

$$\frac{1}{f} = (1.5-1)\left(\frac{1}{15} + \frac{1}{20}\right)$$

This will also result in, f = 17.14 *cm*

Thus, it is concluded that the focal length of the lens will not change if it is flipped by the side. This is true for any lens. Students can verify this for any kind of lens.

EXAMPLE 6.14

Determine the focal length of the lens made up of a material of refractive index 1.52 as shown in the diagram. (Points C_1

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